IGCSE 0607 Practice Questions — E10.5 — Combined Questions & Markscheme

Averages on a calculator

Total Questions: 3 — **Total Marks:** 17 Marks

Table of Contents & Quick Navigation

- **E10.5** Averages on a calculator
  → **Paper 4 (Calculator)** *(3 Qs)*

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1 A survey was conducted to find the number of books read by 21 students during a month.
The results are shown in the table.

Number of books	1	2	3	4	5
Frequency	2	5	7	4	3

(a) Use a graphic display calculator to find the following statistics for this data.

(i) Calculate the mean number of books read.
Give your answer correct to 3 significant figures.

Mean = ... **[2]**

(ii) Find the median.

Median = ... **[1]**

(iii) Find the interquartile range.

Interquartile range = ... **[2]**

2 The table shows information about the time, t minutes, taken by 100 people to travel to work.

Time (t minutes)	Frequency
$0 < t \leq 10$	14
$10 < t \leq 20$	28
$20 < t \leq 30$	35
$30 < t \leq 40$	18
$40 < t \leq 60$	5

(a) Write down the modal class.

..... **[1]**

(b) Use your graphic display calculator to calculate an estimate of the mean time.

..... minutes **[3]**

(c) Explain why your answer to part (b) is an estimate.

..... **[1]**

3 A game is played where players can score $1, 2, 3, 4$, or 5 points.
The table shows the frequencies of the scores obtained by a group of players.
The mean score is 2.8.

Score	1	2	3	4	5
Frequency	10	k	$2k+3$	7	5

(a) Show algebraically that $k = 5$.

[4]

(b) Use your graphic display calculator to find the following statistics for the scores.

(i) Find the median.

Median = ... **[1]**

(ii) Find the lower quartile.

Lower quartile = ... **[1]**

(iii) Find the upper quartile.

Upper quartile = ... **[1]**

1 **(a)** **(i)** 3.0476... **[M1]**

3.05 **[A1]**

(ii) 3 **[B1]**

(iii) Upper quartile = 4 and Lower quartile = 2 identified **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** $20 < t \leq 30$ **[B1]**

(b) Use of correct midpoints: $5, 15, 25, 35, 50$ **[M1]**

$\Sigma fx = 14(5) + 28(15) + 35(25) + 18(35) + 5(50)$ or 2245 seen **[M1]**

22.45 **[A1]**

(c) The exact times are not known so the midpoints of the intervals are used. **[B1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** Total frequency $= 10 + k + 2k + 3 + 7 + 5 = 3k + 25$ **[M1]**

Total score $= 10(1) + k(2) + (2k + 3)(3) + 7(4) + 5(5) = 8k + 72$ **[M1]**

$\dfrac{8k + 72}{3k + 25} = 2.8$ **[M1]**

$8k + 72 = 8.4k + 70 \implies 0.4k = 2 \implies k = 5$ **[A1]**

(b) **(i)** 3 **[B1]**

(ii) 1.5 **[B1]**

(iii) 4 **[B1]**

[Back to Question] *[Total Marks: 7]*